Visualizing genus zero dessins

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Intro

Background

- Number theory, computation

Past

- Postdoc (ICERM)
- Mathematician (CRREL)
  - Computer vision, mixed models

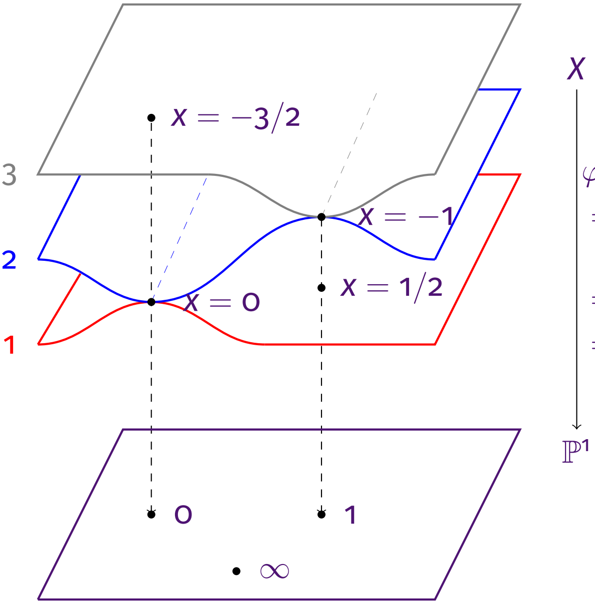
Current

- Data wrangler (Faraday)
  - Product implementation
  - Modeling / API infrastructure
  - Client solutions
- Toddler wrangler

Outline

1. Background
2. Visualizations and objectives
3. Examples

A branched covering



$$\varphi(x) = \underbrace{2x^3 + 3x^2}_{z} = x^2(2x+3) = (2x-1)(x+1)^2 + 1$$

Belyi maps, dessins, etc.

- This branched covering is an example of a **Belyi map**, a map of Riemann surfaces unbranched outside of $0, 1, \infty$.
- A Belyi map determines a **permutation triple** via monodromy.
- A permutation triple determines a **triangle subgroup** Γ which acts on a spherical, Euclidean, or hyperbolic space $\mathcal{H}$ such that $\Gamma\backslash\mathcal{H} \to \Delta\backslash\mathcal{H}$ is the corresponding Belyi map.
- The preimage of $[0, 1]$ is a (connected) bipartite graph embedded on a surface called a **dessin d'enfant**.

Equivalent categories

$$
\begin{tikzcd}
\left\{\parbox{6cm}{Transitive permutation triples $(\sigma_0, \sigma_1, \sigma_\infty) \in S_d^3$ with $\sigma_\infty \sigma_1 \sigma_0 = 1$ up to simultaneous conjugation}\right\} & & \left\{\parbox{5cm}{Conjugacy classes of index d triangle subgroups $\Gamma \leq \Delta(a,b,c)$}\right\} \\
& \left\{\parbox{5cm}{Isomorphism classes of dessins with d edges}\right\} \arrow[ul, leftrightarrow, "\sim"'] \arrow[ur, leftrightarrow, "\sim"] \arrow[dl, leftrightarrow, "\sim"'] \arrow[dr, leftrightarrow, "\sim"] & \\
\left\{\parbox{7cm}{Isomorphism classes of degree d function field extensions $K \supseteq \mathbb{C}(z)$ with discriminant supported above $0, 1, \infty$}\right\} & & \left\{\parbox{5cm}{Isomorphism classes of Belyi maps of degree d}\right\}
\end{tikzcd}
$$

Degree 3 example

$$\big((1\,2)(3), (1)(2\,3), (1\,3\,2)\big)$$

$$[\Delta(2,2,3) : \Gamma] = 3$$

$$\mathbb{C}\left(\underbrace{2x^3+3x^2}_{z}\right) \subseteq \mathbb{C}(x)$$

$$\varphi(x) = \underbrace{2x^3+3x^2}_{z}$$

Galois action

Theorem (Belyi, 1979)
A smooth projective curve X over $\mathbb{C}$ can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a nonconstant morphism of algebraic curves $\varphi : X \to \mathbb{P}^1$ unramified outside $\{0, 1, \infty\}$.

In the 1980s Grothendieck described an action of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the set of (isomorphism classes of) dessins.

Passports

A **passport** consists of the data (G, λ) where

- $G \leq S_d$ is a transitive subgroup
- $\lambda = (\lambda_0, \lambda_1, \lambda_\infty)$ is a triple of partitions of d

A permutation triple $\sigma \in S_d^3$ **belongs** to (G, λ) if

- $\langle \sigma_0, \sigma_1, \sigma_\infty \rangle = G$
- $\sigma_0, \sigma_1, \sigma_\infty$ have the cycle types specified by $\lambda_0, \lambda_1, \lambda_\infty$

Note

- The **size** of a passport is the number of permutation triples belonging to it (up to simultaneous conjugacy).
- Each passport is stable under the Galois action and therefore is a union of Galois orbits.

Objective of this work

Draw topologically correct dessins to help visualize the Galois orbits of all (genus zero) passports in the LMFDB.

- `https://www.lmfdb.org/Belyi/`
- https://michaelmusty.github.io/dessins/

7T6-5.1.1_4.2.1_3.2.2

- `https://www.lmfdb.org/Belyi/7T6/5.1.1/4.2.1/3.2.2/`
- `https://michaelmusty.github.io/dessins/passports/7T6-5.1.1_4.2.1_3.2.2/index.html`
- All drawings together

5T4-5_5_3.1.1

- https://www.lmfdb.org/Belyi/5T4/5/5/3.1.1/
- https://michaelmusty.github.io/dessins/passports/5T4-5_5_3.1.1/index.html
- All drawings together

9T34-6.1.1.1_5.2.2_6.1.1.1

- `https://www.lmfdb.org/Belyi/9T34/6.1.1.1/5.2.2/6.1.1.1/`
- `https://michaelmusty.github.io/dessins/passports/9T34-6.1.1.1_5.2.2_6.1.1.1/index.html`
- All drawings together

Thanks for listening!